Paper 2

1 (a) (i) Taking upwards as positive,

$v_y = 25\sin 45°$

$v_y \approx 17.7 \text{ m s}^{-1}$

(ii) $v_y{}^2 = u_y{}^2 + 2as_y$

At maximum height, $v_y = 0$.

$0 = (25\sin 45°)^2 + 2(-9.81)s_y$

$s_y \approx 16 \text{ m}$ (shown)

(iii) 1. At height 16 m,

$$K.E. = \frac{1}{2}m(u\cos 45°)^2$$

$$K.E. = \frac{1}{2}mu^2 \times 0.5$$

$$K.E. = 0.5K$$

loss in *K.E.* = gain in *G.P.E.*

$K - 0.5K = G.P.E._{\text{at}16\text{m}}$

$G.P.E._{\text{at}16\text{m}} = 0.5K$

2. At height 8.0 m,

$$G.P.E._{\text{at}8\text{m}} = \frac{1}{2}G.P.E._{\text{at}16\text{m}}$$

$G.P.E._{\text{at}8\text{m}} = 0.25K$

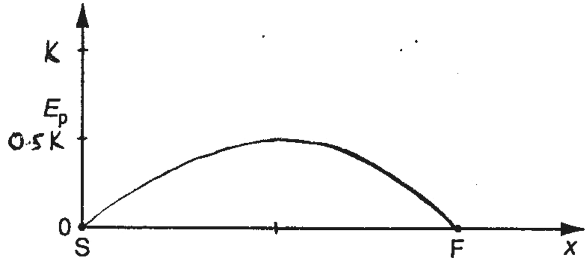
loss in *K.E.* = gain in *G.P.E.*

$K - K.E._{\text{at}8.0\text{m}} = 0.25K$

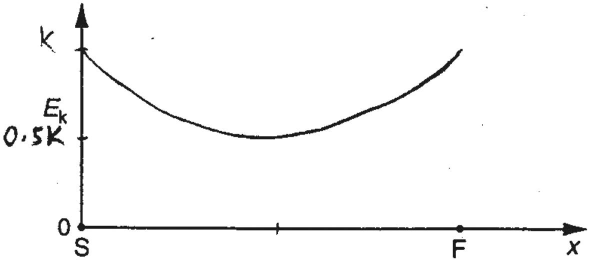
$K.E._{\text{at}8.0\text{m}} = 0.75K$

Note: Candidates should read the question carefully and put their answers in terms of *K*.

(b) (i)



(ii)



2 (a) Insulators have large band gap with a completely filled valence band and empty